

Section One: Multiple Choice

5 marks

Attempt Questions 1-5

Allow about 10 minutes for this section

Write your answer on the paper provided.

1. What are the solutions to $3x^2 - 7x - 1 = 0$?

(A) $x = \frac{-7 \pm \sqrt{37}}{6}$

(B) $x = \frac{-7 \pm \sqrt{61}}{6}$

(C) $x = \frac{7 \pm \sqrt{37}}{6}$

(D) $x = \frac{7 \pm \sqrt{61}}{6}$.

2. Which of the following is irrational?

(A) $\sqrt{289}$

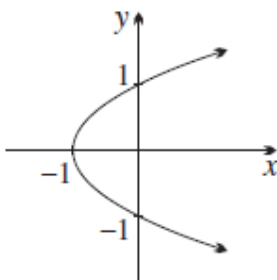
(B) $\sqrt[3]{27}$

(C) $\sqrt{11}$

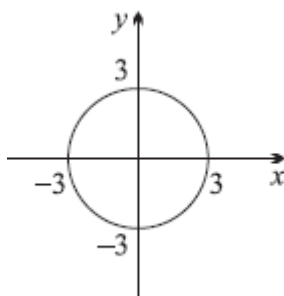
(D) $\frac{3}{7}$.

3. Which of the following is function?

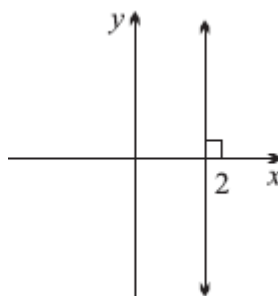
(A)



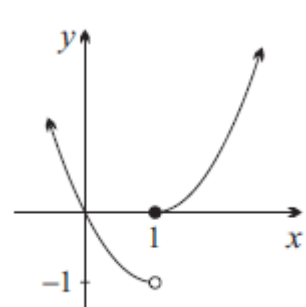
(B)



(C)



(D)



4. Which of the following is true about the function $f(x) = 2^x + 2^{-x}$

(A) Even function

(B) Odd function

(C) Zero function

(D) Neither odd or even function.

5. Simplify $\sin \theta \sec \theta$.
- (A) $\operatorname{cosec} \theta$
- (B) $\sin^2 \theta$
- (C) $\tan \theta$
- (D) $\cot \theta$.

Section Two: Extended Response

Attempt Questions 1-5

Write your answer on the paper provided.

Show all working.

Question One: Methods in Algebra (13 Marks)

Marks

- a) Expand and simplify: $3(2x-1)-3(x+5)$. **2**
- b) Fully factorise: $x^3+5x^2-9x-45$. **2**
- c) Express $\frac{2x-3}{6} - \frac{x-7}{15}$ as a single algebraic fraction. **3**
- d) Solve the equation : $\frac{x-2}{x+2} = 2-x$. **3**
- e) Fully simplify $\frac{x^3-64}{4-x} \div \frac{x^2+4x+16}{5}$. **3**

Question Two: Numbers and Surds (13 Marks)

Marks

Start a new page

a) Fully simplify the following expressions:

i) $\sqrt{27} \times \frac{1}{3}\sqrt{3}$ 2

ii) $(2\sqrt{5} + \sqrt{3})(7\sqrt{3} + \sqrt{5})$ 2

iii) $\sqrt{x^3} + \sqrt{x} - \sqrt{4x}$ 2

b) Express $0.\dot{2}\dot{7}$ as a fraction in its simplest form without the use of a calculator. 2

c) Simplify the following by rationalising the denominator $\frac{\sqrt{5}}{4\sqrt{5} + 7}$. 3

d) Given that a, b, c and d are integers, with b and d non-zero, 2
simplify the average of $\frac{a}{b}$ and $\frac{c}{d}$.

Question Three: Functions and Graphs (13 Marks)**Mark****Start a new page**

- a) If $f(x) = x^2 - 3x - 40$ find $f(5) + f(-5)$ 2
- b) Draw separate neat sketches, labelling significant features in the following:
- i) $xy = 3$ 2
- ii) $y = x^2 + 3x - 10$ 2
- c) State the domain and range for b) i) and b) ii). 4
- d) A function is defined by the following: 3
- $$f(x) = \begin{cases} \sqrt{9-x^2} & \text{for } -3 < x < 3 \\ x & \text{for } x \geq 3. \end{cases}$$

Copy and complete the table of values for $f(x)$, then sketch its graph.

x	-4	-3		-2	-1	0	1	2		3	4
y											

Question Four: Graphs and Inequations (13 Marks)**Marks****Start a new page**

- a) Evaluate $\left| (-3)^2 - 7 \times 2 \right|$. **1**
- b) Solve for x given the inequation $19 < 7x + 5 < 75$. **2**
- c) Solve $|2x + 1| \leq 9$. **2**
- d) Solve the inequality $x(x + 7)(5 - x) < 0$. **3**
- e) On a number plane, graph the following regions: **5**
 $(x - 1)^2 + y^2 \leq 16$ and $x - y + 3 \geq 0$

And shade the intersection of the regions.

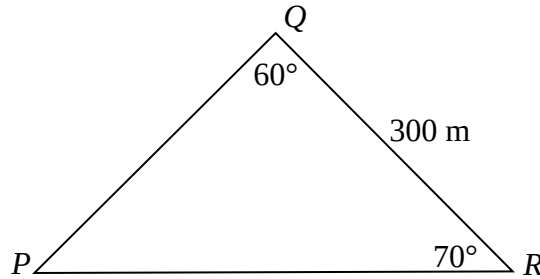
Question Five: Trigonometry (13 Marks)

Marks

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a) If $0^\circ \leq \theta \leq 360^\circ$ find the values of θ given $\sec \theta = \frac{2}{\sqrt{3}}$ **3**

b) In the diagram below, use the Sine rule to find the length of PQ to the nearest metre. **3**



NOT TO SCALE

c) Given that the sides of a non right angled triangle are 6 cm, 11 cm and 15 cm:

i) Show that the smallest angle in the triangle is 20° to the nearest degrees. **2**

ii) Find the area of the triangle, give your answer to 1 decimal place. **2**

d) Prove that $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta - \sin^2 \theta$. **3**

END OF EXAMINATION

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Year 11 - 2 unit Mathematics Half Yearly Examination solutions 2015. <u>Multiple Choice</u> Q1. $a = 3$ $b = -7$ $c = -1$ $x = \frac{-(-7) \pm \sqrt{(-7)^2 - (4 \times 3 \times -1)}}{2 \times 3}$ $x = \frac{7 \pm \sqrt{49 - -12}}{6}$ $x = \frac{7 \pm \sqrt{61}}{6} \quad (D)$ Q2. Rational numbers can be expressed as a fraction. $\sqrt{289} = \frac{17}{1} \quad 3\sqrt{27} = \frac{3}{1}$ $\therefore \sqrt{11}$ is the only value which can not be expressed as a fraction. (C) Q3 Use the vertical line test. A function has one y value for every x value. (D)	Take Care with negative numbers

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q4. $f(-x) = 2^{-x} + 2^{-(-x)}$ $= 2^{-x} + 2^x$ $= f(x)$ When $f(x) = f(-x)$ it is an even function (A) Q5 $\sin \theta \times \frac{1}{\cos \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta$ (C)	The multiple choice questions were generally answered very well by students.
Extended Response Question One. a) $3(2x-1) - 3(x+5)$ $= 6x - 3 - 3x - 15$ ① $= 3x - 18$ ① b) $x^3 + 5x^2 - 9x - 45$ $x^2(x+5) - 9(x+5)$ $(x^2-9)(x+5)$ ① $(x+3)(x-3)(x+5)$ ①	Question One is Reference to Ch1 of textbook. Recognise Grouping and difference of 2 Squares.

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
c) $\frac{2x-3}{6} - \frac{x-7}{15}$ $= \frac{5(2x-3) - 2(x-7)}{30}$ ① $= \frac{10x-15-2x+14}{30}$ ① $= \frac{8x-1}{30}$ ①	Some students failed to successfully change the fractions to have a LCD of 30. 'ect' marks awarded if correct simplifying after failed attempt at changing fractions.
d) $\frac{x-2}{x+2} = \frac{2-x}{1}$ $x \neq -2$ $x-2 = (2-x)(x+2)$ ① $x-2 = 4-x^2$ $x^2+x-6=0$ ① $(x+3)(x-2)=0$ $x=-3 \quad x=2$ ①	Recognise $(2-x)(x+2)$ $= (2-x)(2+x)$ Difference of 2 squares. some students struggled to get to this step by eliminating the $x+2$ denominator from the equation.
e) ① $\frac{(x-4)(x^2+4x+16)}{(4-x)} \div \frac{(x^2+4x+16)}{5}$ $= \frac{(x-4)(x^2+4x+16)}{(4-x)} \times \frac{5}{(x^2+4x+16)}$ ① $= -1 \times 5$ $= -5$ ①	since $(4-x) = -1(x-4)$ some students forgot diff. 2 cubes factorisation.

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)

comments

Extended Response Question Two.

$$a) \quad (i) \quad \sqrt{27} \times \frac{1}{3}\sqrt{3}$$

$$= 3\sqrt{3} \times \frac{1}{3}\sqrt{3} \quad (1)$$

$$= 3 \quad (1)$$

$$(ii) \quad E. \quad 0. \quad 1. \quad L$$

$$14\sqrt{15} + 10 + 21 + \sqrt{15} \quad (1)$$

$$15\sqrt{15} + 31 \quad (1)$$

$$(iii) \quad x\sqrt{x} + \sqrt{x} - 2\sqrt{x} \quad (1)$$

$$= \sqrt{x}(x-1) \quad (1)$$

$$b) \quad \text{Let } x = 0.277777 \dots$$

$$10x = 2.777777 \dots$$

$$100x = 27.777777 \dots \quad (1)$$

$$90x = 25$$

$$x = \frac{25}{90} = \frac{5}{18} \quad (1)$$

$$c) \quad \frac{\sqrt{5}}{4\sqrt{5}+7} \times \frac{4\sqrt{5}-7}{4\sqrt{5}-7} \quad (1)$$

$$\frac{20-7\sqrt{5}}{80-49} \quad (1)$$

$$= \frac{20-7\sqrt{5}}{31} \quad (1)$$

ok ✓

generally ok ✓

many didn't factorise

Some use their calculator! Ooops!

Some x by $4\sqrt{5}+7$ and some left out +

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)

$$d) \frac{\frac{a}{b} + \frac{c}{d}}{2}$$

$$\frac{ad+bc}{bd}$$

$$\frac{\quad}{2}$$

$$\frac{ad+bc}{2bd}$$

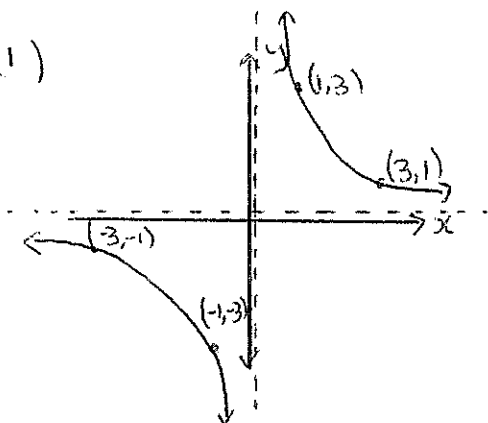
Extend Response Question 3

$$a) f(s) = s^2 - 3(s) - 40 = -30$$

$$f(-s) = (-s)^2 - 3(-s) - 40 = 0 \quad (1)$$

$$f(s) + f(-s) = -30 \quad (1)$$

b) (1)



comments

Many didn't attempt this question.

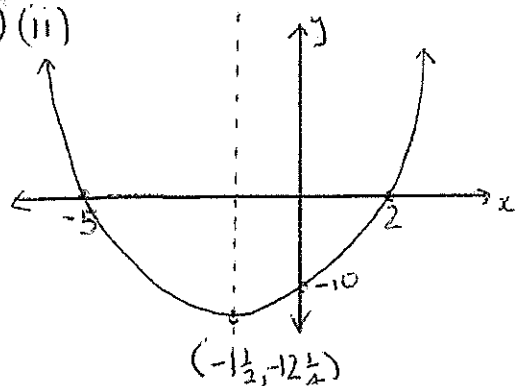
✓

many omitted asymptotes
• very poor sketch

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)

b) (ii)



$$y = (x+5)(x-2)$$

(2)

comments

• poor sketch
need a single
smooth curve

• vertex not shown

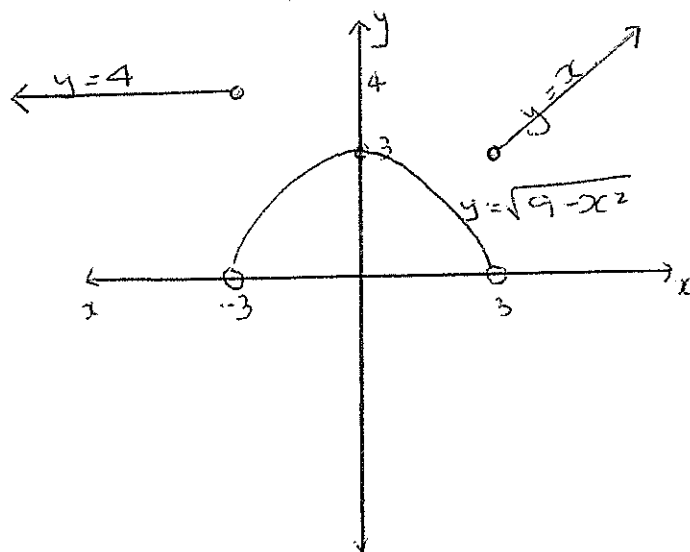
c) b(i) domain: all real x exceptwhen $x = 0$ ①range: all real y exceptwhen $y = 0$ ①b(ii) domain: all real x ①range: all real $y \geq -12\frac{1}{4}$

①

omitted $x \neq 0$ $y \neq 0$

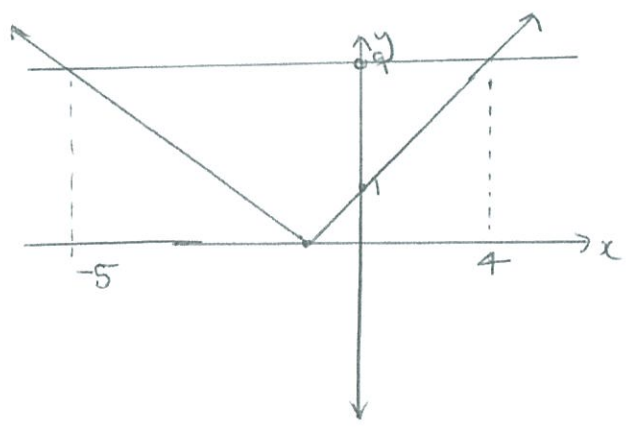
d)

x	-4	-3	-2	-1	0	1	2	3	4
y	4	4	$\sqrt{5}$	$\sqrt{8}$	3	$\sqrt{8}$	$\sqrt{5}$	3	4

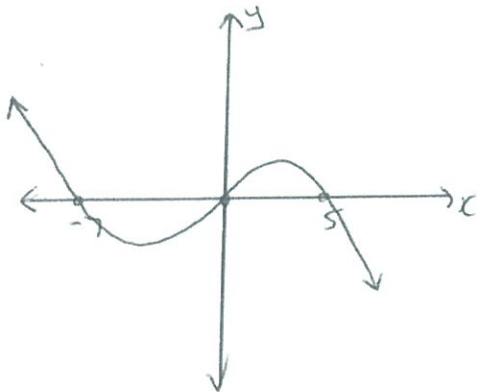
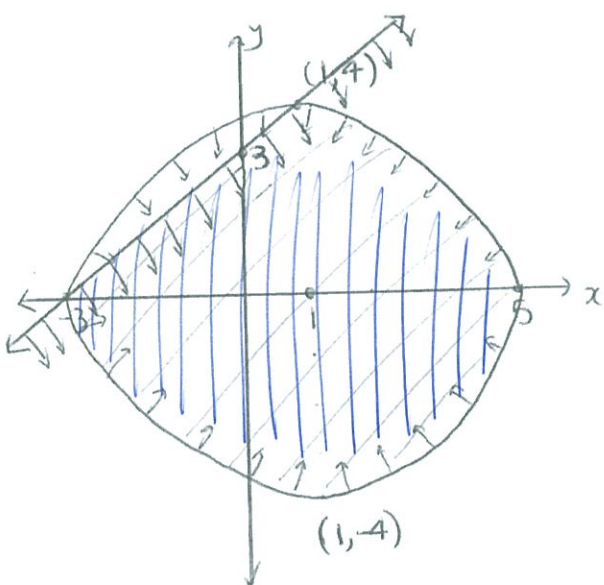


very poorly
drawn.

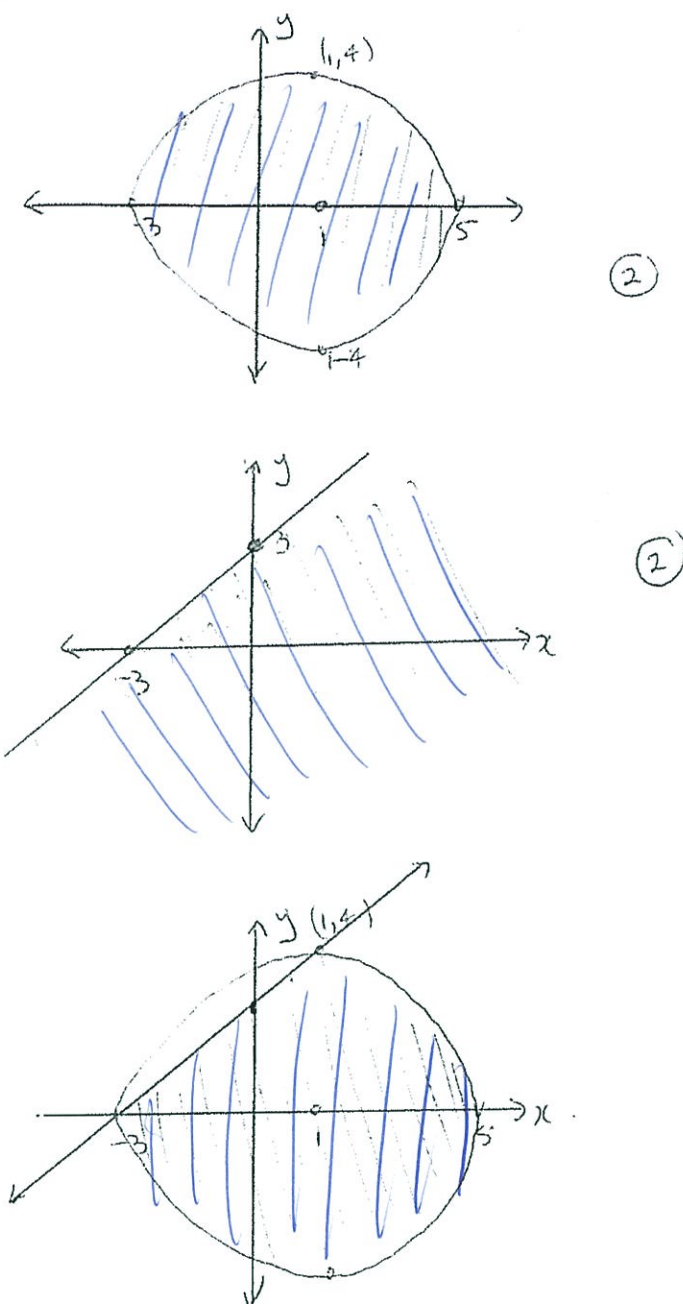
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Extended Response Question Four a) $9-14 = -5 = 5$ ① b) $19 < 7x+5 < 75$ $14 < 7x < 70$ ① subtract 5 $2 < x < 10$ ① divide 7 c) $2x+1 \leq 9$ $2x+1 \leq 9$ $2x+1 \geq -9$ $2x \leq 8$ $2x \geq -10$ $x \leq 4$ $x \geq -5$ ① $-5 \leq x \leq 4$ ① check solution makes sense. 	only one answer = 5 -5 and -5 was NOT correct and written in addition to 5 meant no mark. often only one solution was given.

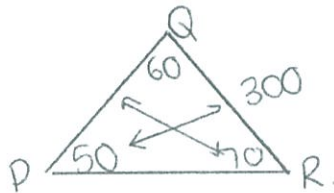
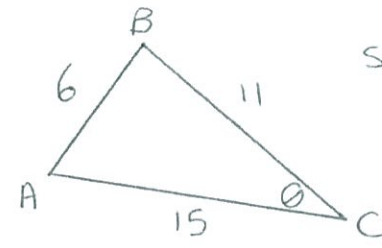
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
d) $x(x+7)(5-x) < 0$ zeroes occur when $x=0$ $x=-7$ $x=5$ When $x \rightarrow \infty$ $y \rightarrow -\infty$  below y-axis $-7 < x < 0$ $x > 5$	1 mark for (3) value $x=0$ $x=-7$ $x=5$ 1 mark for graph/other. If graphed incorrectly e. c. f. was awarded for final mark.
e)  2 marks for graph of circle and correct region. 2 marks for graph of line and correct region. 1 mark for correct intersection.	Many students did not test a point and shaded the incorrect region (see over page)

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u>	<u>comments</u>
(e) 	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Question Five . a) $\frac{1}{\cos \theta} = \sec \theta$ $\therefore \cos \theta = \frac{\sqrt{3}}{2} \quad (1)$ $\theta = 30 \quad (1) \qquad \theta = 330 \quad (1)$	Good .
b)  $\angle QPR = 50^\circ \quad (1)$ $\frac{PQ}{\sin 70} = \frac{300}{\sin 50}$ $PQ = \frac{300 \sin 70}{\sin 50} = 368 \text{ m}$	Must use Sine Rule .
c)  smallest angle is opposite smallest side $\cos \theta = \frac{15^2 + 11^2 - 6^2}{2 \times 15 \times 11} \quad (1)$ $\cos^{-1}(\text{ANS})$ (shift) $\cos \theta = 20^\circ \quad (1)$	

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u>	<u>comments</u>
C (ii) $\text{Area} = \frac{1}{2} ab \sin C$ $\text{Area} = \frac{1}{2} \times 11 \times 15 \times \sin 20$ $\text{Area} = 28.2 \text{ cm}^2$ d) $\text{LHS} = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ since $\tan^2 \theta + 1 = \sec^2 \theta$ $\text{LHS} = \frac{1 - \tan^2 \theta}{\sec^2 \theta} \quad (1)$ $= \frac{1}{\sec^2 \theta} - \frac{\tan^2 \theta}{\sec^2 \theta}$ $= \cos^2 \theta - \frac{\sin^2 \theta}{\cos^2 \theta} \div \sec^2 \theta$ $= \cos^2 \theta - \frac{\sin^2 \theta}{\cos^4 \theta} \times \cos^2 \theta$ $= \cos^2 \theta - \sin^2 \theta$ $= \text{RHS.}$ $\therefore \text{LHS} = \text{RHS.}$ $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta - \sin^2 \theta$	There is No Right Angle in the triangle. use $\text{LHS} = \text{RHS}$